Text-to-Cypher: An Agentic Approach

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Abstract

The translation of natural language questions into database queries, known as Text-to-Query, remains a significant challenge, especially for complex formalisms like Cypher used in Property Graph Databases. While recent advancements have leveraged Large Language Models (LLMs), existing solutions often rely on extensive fine-tuning or fail to adequately address the semantic gap between unstructured intent and rigid schema structures. In this paper, we propose a novel, zero-shot agentic framework for the Text-to-Cypher task that autonomously navigates the schema discovery and query generation process. Our system, built on LangChain and LangGraph with GPT-4.1 as the core reasoning engine, employs a specialized suite of tools and a hybrid indexing strategy combining vector-based semantic retrieval with structural inverted indexes. This architecture enables the agent to systematically identify key concepts, discover relevant formalisms, link them effectively, and validate assumptions through direct database interaction. We evaluate our approach on the *neo4j/text2cypher-2024v1* benchmark, demonstrating that our zero-shot agent matches the performance of state-of-the-art fine-tuned models, achieving an Exact Match score of 32.62%. Furthermore, we provide a detailed analysis of the agent's behavior, highlighting its highly parallelized "search-then-refine" strategy and identifying current limitations in handling complex, non-linear reasoning paths. Our findings suggest that general-purpose reasoning agents, equipped with domain-specific tools, offer a powerful and flexible alternative to traditional fine-tuning for semantic parsing tasks.

Keywords

Text-to-Query, Text-to-Cypher, Information Retrieval, Knowledge Graphs, Agentic Systems

1. Introduction

For decades, the problem of bridging the gap between natural language and formal languages had always been a significant challenge, both for humans and software agents, in the field of computer science. One instance of this challenge is the task of converting user's questions into database queries, with the main goal of retrieving relevant information from *structured* data sources starting from *unstructured* user queries expressed in natural language. This class of tasks, commonly referred to as Text-to-Query, is fundamentally a challenge of semantic translation and alignment between two distinct representational paradigms: the often ambiguous nature of human language and the rigid, formal structure of query languages.

The proliferation of large-scale Knowledge Graphs (KGs) has revolutionized how structured information is stored and integrated across domains ranging from biomedicine to financial intelligence. Among these, Property Graph Databases, most notably Neo4j ones with their declarative query language Cypher, have emerged as the industry standard for representing and querying complex, interconnected data. Text-to-Cypher, a specialized subset of the broader Text-to-Query task, focuses on translating natural language questions into Cypher queries that can be executed against Property Graph Databases. This task is particularly challenging due to the inherent complexity of graph structures to express and the way in which Cypher allow us to represent such complex graph patterns (that requires some vision capabilities and intelligence to better interpret and understand ASCII art styled cypher queries).

To solve this class of tasks, an intelligent agent must be able to perform, at least, the following sub-tasks:

1. **Relevant Concepts Identification**: extracting key *entities*, *relationships* between them and their *properties* of our interest expressed in natural language. This process is the most problematic

Woodstock'22: Symposium on the irreproducible science, June 07–11, 2022, Woodstock, NY
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one since it is often carried out by abstract innate human cognitive processes, guided by specific intents, which are difficult to formalize computationally and requires advanced natural language understanding with a deep world knowledge.

2. **Relevant Formalisms Discovery**: discoverying the appropriate formal constructs, of the underlying used formalism framework, that had or could be used to represent identified concepts. This step involves understanding the syntax and semantics of this symbolical framework by performing multiple reasoning and action steps. In our case, the underlying framework is represented by the Property Graph Data Model and the Cypher formal query language. In our case, the discovery process involves exploring the target graph database schema to find relevant *node labels*, *relationship types* and their *properties* that had or can be used to represent identified concepts.
3. **Concepts-Formalisms Linking**: establishing correspondences between identified concepts and discovered formal constructs. In our case, some instances of this sub-task can be referred to as **Entity Linking**, **Relationship Linking** and **Property Linking**.
4. **Formalisms Framework Exploitation**: leveraging the established correspondences to construct syntactically and semantically correct queries in the target formalism. In our case, this involves generating Cypher queries that accurately represent the user's natural language questions based on the linked concepts and formal constructs.

All these sub-tasks are inherently probabilistic or possibilistic in nature and require a deep understanding of natural language, general world knowledge and the target formalism, as well as the ability to reason about relationships between concepts and their representations. With the advent of Large Language Models (LLMs) and their agentic capabilities, new avenues had opened for addressing the inherent challenges in this task. Their proficiency in understanding natural language, reasoning over complex information and generating coherent text to perform specific actions, makes them well-suited for performing the sub-tasks outlined above.

In this paper, we propose an agentic solution to Text-to-Cypher tasks, without any kind of fine-tuning, that leverages LLMs to first find all relevant schema components in the target graph database, necessary for query construction and then generate the final Cypher query to be executed.

Contributions Our core contributions are summarized as follows:

- We propose a novel **agentic framework** for Text-to-Cypher that leverages Large Language Models (LLMs) to distinctively address schema discovery and query generation.
- We design a specialized **toolset and indexing strategy**, combining vector-based semantic retrieval with structural inverted indexes, that enables agents to effectively explore graph schemas and validate assumptions against actual data.
- We provide a comprehensive **evaluation** on the neo4j/text2cypher-2024v1 benchmark, demonstrating that our zero-shot agentic approach matches the performance of fine-tuned state-of-the-art models (32.62% vs 32.50% Exact Match) while offering superior explainability and flexibility.

2. Related Works

The current landscape of Text-to-Query research is marked by a variety of approaches that leverage different techniques and methodologies to address typical problems encountered in this class of tasks. Broadly speaking, we can categorize existing works into two main lines of research: (i) the development of benchmark datasets specifically designed for training and evaluating Text-to-Query models, and (ii) the design of general Text-to-Query solutions that can be adapted to various types of databases and query languages. For our purposes, in the first line of research, we are mainly interested in datasets focused on the Text-to-Cypher task, while in the second line of research, we will explore more general Text-to-Query solutions that can be adapted to our specific task.

2.1. Text-to-Cypher Datasets

An example include the neo4j/text2cypher-2024v1 dataset introduced by the Neo4j-Labs team [1], which brings together instances from publicly available datasets, cleaning and organizing them for smoother use. This dataset consists of 44,387 instances, with 39,554 instances in the training split and 4,833 instances in the test split where for each instance, a natural language question is paired with: its corresponding Cypher query, the entire schema of the underlying graph database and, if available, an identifier to the specific Neo4j database hosting the entire knowledge graph from which the instance was derived. The Neo4j-Labs team, in fact, provide a remote access to multiple already deployed Neo4j graph databases, freely and publicly accessible, that can be used to execute the Cypher queries present in the dataset. This important aspect paired with the vast diversity of the included knowledge graphs (about 17 different Neo4j graph databases are used to derive the instances in the dataset), makes it well-suited for our purposes.

Another notable example is the megagonlabs/cypherbench dataset introduced by Megagon Labs [2] with 10,882 instances for training and evaluating text-to-Cypher models. To create such a benchmark, the authors first developed a novel RDF-to-Property-Graph transformation engine that converts RDF data into high quality locally deployable property knowledge graphs while preserving the original semantics. Then, they applied this engine to Wikidata, generating 11 large-scale multi-domain property graphs that captures a wide range of real-world entities and relationships. Lastly, they designed a synthetic data generation pipeline that, starting from graph patterns templates, automatically creates diverse natural language questions and their corresponding Cypher queries based on the structure and content of the generated property graph. The peculiarity of this approach lies in its focus on comprehensiveness, in graph patterns covered, and diversity, in graph schemas and use cases addressed, of the resulting benchmark dataset. However, the lack of already deployed, publicly accessible and remote Neo4j graph databases endpoints, force anyone willing to use this dataset to first locally deploy the generated property graphs before being able to execute the Cypher queries present in the dataset. To further complicate things, the configuration method of such local deployments provided by the authors requires significant hardware requirements (at least 64 GB of RAM when deploying the 7 test graphs and 128GB when deploying all 11 graphs[1]) that are out of reach for many researchers. For this reason, we decided to not use this dataset for our experiments and to focus on easy-to-use datasets that allow us to quickly experiment novel approaches without the need for complex and resource-intensive local deployments.

2.2. General Text-to-Query Solutions

Several solutions have been proposed to tackle the general Text-to-Query translation task, primarily leveraging transformer-based architectures [3], both decoder-only and encoder-decoder models, in combination with various techniques. To address the problem of **Relevant Concepts Identification** and to ensure the generation of **schema-compliant**, syntactically and semantically correct queries, researchers have extensively adopted **in-context learning** (ICL) combined with **prompt engineering**. These techniques exploit the few-shot and zero-shot capabilities of large language models by providing carefully crafted prompts containing *task instructions*, *few-shot demonstrations* (retrieved from the training set), and relevant contextual information such as *schema elements* to guide query generation [1, 4, 5, 2, 6].

Schema Retrieval Approaches. Regarding the inclusion of schema information in prompts, most existing works [1, 2, 4, 6] either assume that relevant schema components are already known or simply provide the entire database schema without any filtering or pruning. This approach, while straightforward, can lead to excessively large prompts that may exceed the model's maximum context length or introduce irrelevant information that degrades generation quality.

[1]This was explicitely stated in the 3rd step of the quick start of the `README.md` file in their repo https://github.com/megagonlabs/cypherbench

To address problems 1, 2, and 3 in Section 1, only few works have proposed techniques for **targeted schema retrieval**. Some approaches [5, 7, 8] combine lexical matching methods (e.g., Levenshtein distance, fuzzy string matching, token-based or prefix matching) for schema element identification with semantic similarity techniques (e.g., BM25 sparse retrieval or dense embeddings from Sentence Transformers [9]) for instance retrieval.

Trade-offs in Similarity Techniques. These retrieval methods, however, present inherent trade-offs. Lexical matching excels at identifying exact matches or minor spelling variations in short terms but becomes less effective for longer phrases where semantic understanding is crucial (e.g., handling synonymy, polysemy, and other lexical-semantic relationships). Conversely, contextual dense embeddings from transformer models excel at capturing semantic similarity and scale efficiently to large databases through vector stores. However, they can be inefficient or ineffective for short, information-dense terms such as node labels (`DataCenter`, `PointOfInterest`), relationship types (`PARENT_OF`, `DEPENDS_ON`), or property keys (`created_at`, `description`) which lack the rich explicit context needed to generate discriminative contextual representations useful for retrieval purposes.

Static Word Embedding (SWEs) models like FastText [10] address this limitation by providing context-free, semantically rich representations that handle out-of-vocabulary terms through subword information. However, their prohibitive memory requirements (vocabularies with millions of tokens) make them impractical for large-scale deployment.

Model2Vec [11] offers an optimal solution for this use case. It produces static word embeddings by distilling knowledge from large Sentence Transformer models through output embedding aggregation and dimensionality reduction via PCA [12], eliminating the need for the original model's heavy neural architecture. This distillation process yields compact, semantically rich representations that retain the semantic quality of their teacher models while requiring only vocabulary-level storage and enabling inference through simple vocabulary lookups and mean pooling. For our scenario, Model2Vec provides an ideal balance: it captures rich semantic information like contextual models, operates efficiently on short terms like static embeddings, yet avoids the computational overhead of transformer inference and the memory footprint of traditional SWEs.

Algorithmic vs. Agentic Architectures. Text-to-query solutions can be further categorized based on their architectural paradigm. **Algorithmic approaches**, such as Yang et al. [5], follow predefined pipelines: first performing entity linking via lexical matching at the schema level and hybrid lexical-semantic matching at the instance level, then extracting relevant schema portions for query construction. While deterministic, these approaches face several limitations: (i) they rely on strong, domain-specific assumptions that may not generalize, (ii) they lack robust disambiguation mechanisms when multiple candidates match the same concept, and (iii) they treat relationship linking as secondary to entity linking, imposing rigid dependencies that may not hold in real-world scenarios.

In contrast, **agentic approaches** [13, 14, 15], such as the framework proposed by Walter et al. [7, 8], equip LLM agents with a set of tools to iteratively address all sub-tasks 2, 3, and 4 from Section 1. These tools include pre-built indexes for lexical and semantic search, as well as direct database query capabilities for grounding and query formulation. This architecture enables dynamic tool selection based on context, allowing the agent to adapt its retrieval strategy to specific database characteristics and handle ambiguities more flexibly than rigid algorithmic pipelines. However, ensuring efficiency and effectiveness at scale, particularly for large databases or complex queries, remains an open challenge requiring careful system design and optimization.

3. Approach

In this section, we present our agentic solution for the Text-to-Cypher task. We first describe the overall system architecture, highlighting the core components and their interactions. Then, we detail the design and implementation of the specific tools that enable our agent to effectively perform all the sub-tasks defined in Section 1.

3.1. System Architecture

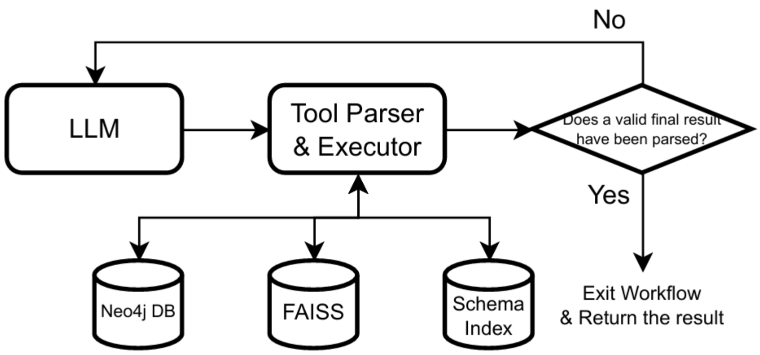


Figure 1: System Architecture

Our system is built upon the LangChain[2] and LangGraph[3] frameworks, which provide robust infrastructure for implementing stateful workflows. The architecture employs OpenAI's GPT-4.1 as the central reasoning engine, responsible for interpreting user queries and orchestrating the execution of sub-tasks necessary to generate the final Cypher query.

To enable **Relevant Formalisms Discovery**, the system incorporates a **FAISS Vector Store** [16] for semantic retrieval of schema components. This vector store indexes node labels, relationship types, and property keys from the target knowledge graph using the minishlab/potion-retrieval-32M Model2Vec embedding model, which is a finetuned version of the minishlab/potion-base-32M[4] model

To capture the structural relationships between schema components, we designed a complementary **Schema Index** i.e. an inverted index data structure that maintains bidirectional mappings between: (i) node labels and their properties, (ii) relationship types and their properties, (iii) relationship types and their valid domain-range node label pairs, (iv) property keys and their associated node labels, and (v) property keys and their associated relationship types. This index enables efficient reverse lookups and schema traversal queries.

The system operates through a ReAct-style agentic loop [13], showed in figure 1, iteratively invoking tools and reasoning over observations until (i) the agent either accumulates sufficient information to construct a valid Cypher query or (ii) determines that the user's question cannot be answered given the available knowledge graph or (iii) reaches a predefined graph recursion limit (25: the LangGraph default one in our case). To ensure effective agentic behavior, we employ several prompting techniques: the system prompt enforces a *Zero-Hallucination Policy*, explicitly instructing the agent that it has no built-in schema knowledge and must verify every component through tool calls; it guides *structured question decomposition* along entities, relationships, attributes, and constraints dimensions with concrete candidate generation examples (e.g., "movies" $\rightarrow$ {movie, film, production, media}); and it emphasizes *batching and parallelism patterns* to minimize token consumption, with tool docstrings programmatically extended via markdown files containing extensive few-shot examples demonstrating proper batch operations and reasoning traces effectively exploiting In-Context Learning capabilities of the LLM.

[2] https://www.langchain.com/
[3] https://www.langchain.com/langgraph
[4] Which in turn it is a Model2Vec model distilled from the baai/bge-base-en-v1.5 Sentence Transformer [17]

3.2. Tool Design and Implementation

To enable the LLM agent to systematically perform the sub-tasks 2 and 3 outlined in Section 1, we designed a suite of specialized tools that facilitate efficient and effective schema exploration and information retrieval.

```
def search_for(
    components: List[Literal["nodes", "relationships", "properties"]],
    similar_to: str,
    top_k: int = 10
) -> str:
```

The `search_for` tool performs semantic search over specified FAISS-indexed schema components (node labels, relationship types, and property keys) to identify all plausible candidate formalisms that could represent a given concept. The tool accepts three parameters: `components` (specifying which schema component types are included in the search space), `similar_to` (a natural language description of the concept), and `top_k` (the number of most similar results to return). This design explicitly prioritizes *recall over precision*: the tool is intended to cast a wide net and retrieve all potentially relevant schema components, even at the cost of including some false positives. This design choice is crucial because missing a relevant schema component early in the retrieval process would be more costly to recover in subsequent steps, whereas false positives can be filtered out through later precision refinement, delegating it to subsequent tools. To maximize efficiency and minimizing latency, the tool is designed to be invoked in parallel for each independent concept within a complex query. Lastly, both here and in the subsequent tools, `similar_to` and `top_k` parameters are always used to limit results only to the most relevant one avoiding polluting and consuming the LLM token context window with useless information.

```
def get_properties_from(
    components: List[Literal["nodes", "relationships"]],
    of_types: List[str],
    similar_to: Optional[List[str]] = None,
    top_k: Union[int, List[int]] = 10
) -> str:
```

Once relevant node labels or relationship types have been identified, for each *i-th* element in the batch, the `get_properties_from` tool retrieves the associated properties of the schema component `components[i]` of type `of_types[i]` optionally ranked by similarity to `similar_to[i]`. This *batching-first* design philosophy is critical for two reasons: (i) it reduces the number of LLM tool calls, thereby decreasing latency and token consumption, and (ii) it encourages the agent to think holistically about multiple schema components simultaneously rather than myopically focusing on one at a time.

```
def get_domains_ranges_of(
    relationships: List[str],
    similar_to: Optional[List[str]] = None,
    order_by: List[Literal["domains", "ranges"]] = None,
    top_k: Union[int, List[int]] = 10
) -> str:
```

The `get_domains_ranges_of` tool for each relationship type `relationships[i]` retrieves valid pairs of domain (source) and range (target) node labels most similar to the `similar_to[i]` query. The `order_by[i]` parameter allows the agent to specify whether the ranking should be applied on domains or ranges. This tool is essential for understanding how different entities are interconnected within the graph, enabling the agent to reason about potential traversal paths and relationship chains necessary for constructing complex Cypher queries.

```
def get_components_with_property(
    keys: List[str],
```

```
    components: List[Literal["nodes", "relationships", "both"]],
    similar_to: Optional[List[str]] = None,
    top_k: Union[int, List[int]] = 10
) -> str:
```

While most tools follow a forward lookup pattern (from schema components to properties), `get_components_with_property` performs reverse lookup finding, for each property key `keys[i]`, which schema components `components[i]` most similar to the query `similar_to[i]` possess it. This tool is particularly valuable for disambiguating queries where discriminative properties serve as strong signals for **Relevant Formalisms Discovery** (Sub-task 2).

```
def execute_cypher_query(cypher: str) -> List[dict]:
```

The `execute_cypher_query` tool provides a flexible mechanism for the agent to validate assumptions, explore data instances, and verify the existence of specific patterns through the execution of arbitrary Cypher queries. Unlike the schema-focused tools, this tool operates at the data instance level and, therefore, it is crucial for grounding the agent's schema-level reasoning in actual data instances, enabling it to confirm hypotheses that inform the final query construction.

```
def register_retrieval_result(
    motivation: str,
    cypher_query: Optional[str] = None,
    kg_schema: Optional[KGSchema] = None
) -> RetrievalAgent.Result:
```

The `register_retrieval_result` tool allow the LLM to end the ReAct loop by summarizing the findings with a structured output containing: (i) the `motivation` explaining why the user's query can or cannot be answered with a cypher query given the discovered schema components in the provided knowledge graph, (ii) optionally the constructed `cypher_query`, and (iii) an optional `kg_schema` object that encapsulates the discovered sub-schema relevant to the user's query. This structured output is essential for downstream components that may need to further process or execute the generated Cypher query.

This carefully orchestrated tool ecosystem enables the LLM agent to systematically navigate the complex space of schema exploration, moving from broad conceptual understanding to precise formal query construction while maintaining both comprehensiveness and efficiency.

4. Experiments

In this section, we present the experimental setup and results of our approach. We evaluated our method only on instances of the neo4j/text2cypher-2024v1 test set that have an already deployed Neo4j database available on the Demo endpoint. This reduced the number of test instances from 4,833 to 2471, but it was strictly necessary to, at least, execute our agentic approach against the actual database used for the generation of samples in the test set. We evaluated both **effectiveness** and **efficiency** of our approach. In particular, we used already computed zero-shot and finetuned effectiveness evaluation results reported in the original paper [1] as several baselines to compare our approach with their one.

4.1. Effectiveness Evaluation Metrics

We evaluate our text-to-Cypher approach using runtime-based metrics, following the evaluation protocols established in prior work [1, 2] assessing the semantic equivalence of generated Cypher queries.

Executable Accuracy (EA). This metric measures whether a generated Cypher query executes successfully on the target Neo4j database without raising syntax or runtime errors. A query is considered valid if it can be executed within a specified timeout (120 seconds in our experiments) without encountering exceptions such as `CypherSyntaxError`, `DatabaseError`, `CypherTypeError`, or `ClientError`.

Result Similarity (RS) and Exact Match (EM). To assess semantic equivalence at the data level, we compute the Jaccard similarity between the result sets returned by the predicted and ground truth queries. Specifically, we represent each query result as a list of dictionaries, where each dictionary corresponds to a row in the output. The similarity computation accounts for row ordering when either query contains an `ORDER BY` clause; otherwise, we apply a greedy alignment algorithm that pairs rows from both result sets to maximize overall similarity. For each aligned pair of rows, we convert values to hashable types and compute their Jaccard index. The final **Result Similarity** is the Jaccard coefficient over all (row index, value) pairs across both result sets. This metric captures whether queries retrieve the same data, even when their structural implementations differ. Following [1], a value of RS $= 1$ is considered an **Exact Match**.

Provenance Subgraph Jaccard Similarity (PSJS). Introduced by [2], this metric evaluates the structural overlap of the graph patterns matched by two queries, rather than their final result sets. The provenance subgraph of a query consists of all nodes and relationships traversed during pattern matching, regardless of what is ultimately returned. For each query, the authors [2] extract the `MATCH` and `WHERE` clauses, augment unlabeled nodes and relationships with temporary variables, and construct a Cypher query that returns the element IDs of all matched nodes. The PSJS is then computed as the Jaccard coefficient between the sets of element IDs from the predicted and ground truth provenance subgraphs:

$$\text{PSJS} = \frac{|PS_{\text{pred}} \cap PS_{\text{target}}|}{|PS_{\text{pred}} \cup PS_{\text{target}}|}$$

where PS_{pred} and PS_{target} denote the provenance subgraphs of the predicted and target queries, respectively. This metric is particularly valuable for evaluating whether a model correctly understands the graph structure to be queried, even if the final projection or aggregation differs from the ground truth.

Success Rate (SR) For each test instance, if the execution of the loop raised any exception (e.g. the graph recursion limit was reached), we assigned null values to all the evaluation metrics for that instance. In this way, we are able to clearly distinguish between failed executions and low-quality query generations. We define the Success Rate as the proportion of test instances where the agent successfully completed its execution without exceptions calling them successful sessions or instances.

4.2. Results

Table 1 summarizes the effectiveness of our agentic text-to-Cypher approach compared to the zero-shot and finetuned best baselines from [1]. As shown, our agentic approach match the best Exact Match (EM) score of 32.50% achieved by the finetuned GPT-4o model, while also addressing the sub-problems 1, 2 and 3 left unsolved by baselines authors [1].

Model	Result Similarity	Exact Match	PSJS	Success Rate
GPT-4o (zero-shot)	-	31.73%	-	-
GPT-4o (fine-tuned)	-	32.50%	-	-
GPT-4.1 (Ours)	41.22%	32.62%	69.37%	89.11%

Table 1
Effectiveness Results Comparison with Baselines from [1]

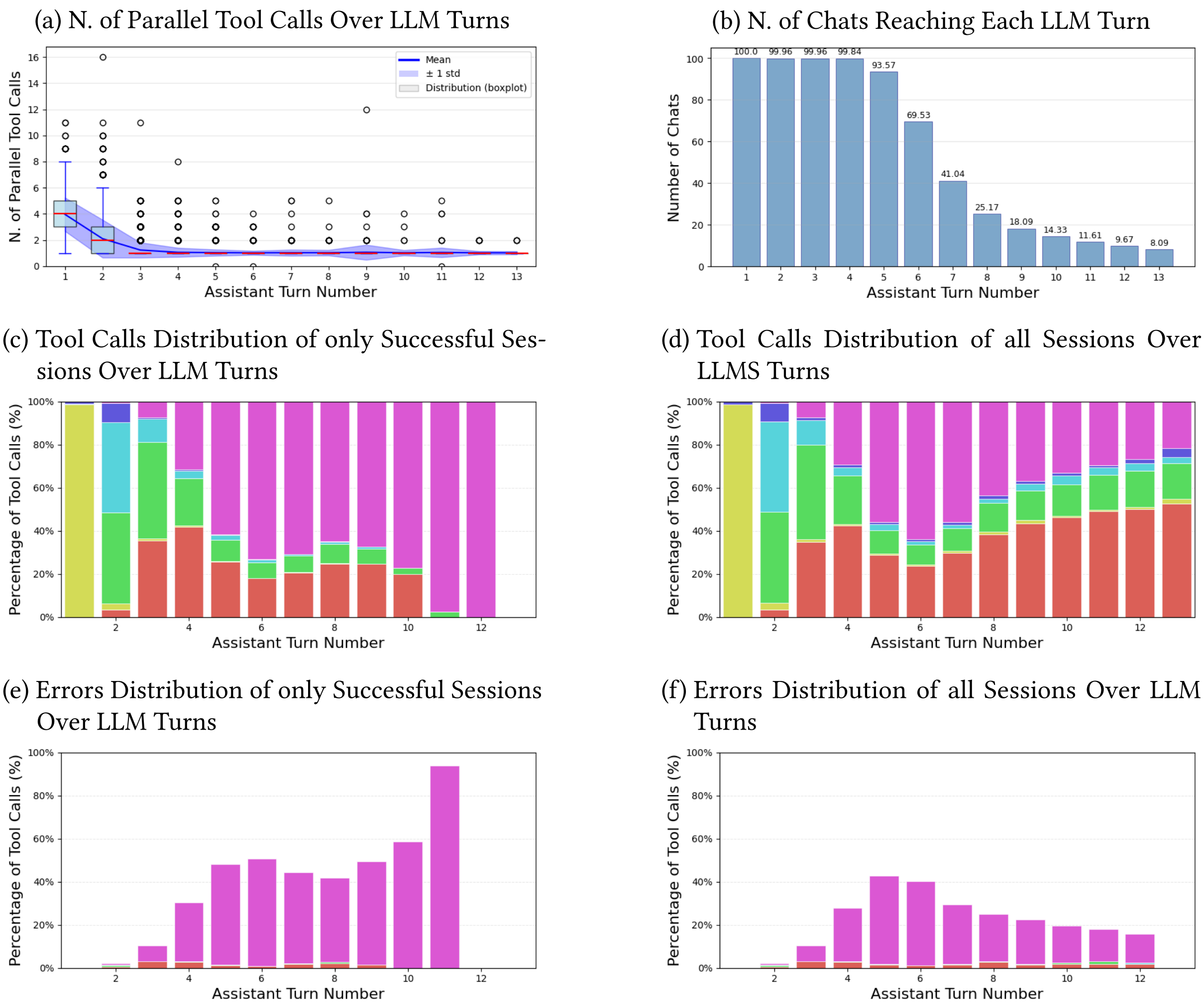
Figure 2 summarizes some efficiency statistics of our approach. As we can observe, from the figure 2a that shows the distribution of the number of parallel tool calls over LLM turns, the moments in which there is the highest parallelism grade are the first 2-3 ones, where the agent exploits the available tools to quickly gather as much information as it can, hopefully reducing the overall entropy of the problem.

The figure 2b shows the number of chats reaching each LLM turn, from the minimum one to the maximum one. We can observe that for the majority of the chats, the agent is able to finish its execution in less than 7-8 LLM turns, with a significant drop after the 5th-6th turn. This is coherent with the Success Rate of 89.11% reported in Table 1, which means that for the majority of the test instances, the agent is able to successfully complete its execution without exceptions.

Figures 2c and 2d (with the color legend included in table 2) show the tool calls distribution over LLM turns respectively without and with the inclusion of failed chat sessions while figures 2e and 2f show the errors distribution over time respectively without and with the inclusion of failed chat sessions. From the first one, we can observe that, for successful sessions, the first turn is completely dominated by `search_for` tool calls, which is coherent with the fact that the agent has no prior knowledge about how identified concepts are represented in the knowledge graph and it needs to perform a broad **Formalisms Discovery** to preserve recall as earliest as possible. As the turns progress, we can observe a more balanced distribution of tool calls, with `get_properties_from` and `get_domains_ranges_of` becoming more prominent. This indicates that as the agent gathers more information about the graph, it starts to focus on specific properties and relationships relevant to the query at hand. The appearance of `execute_cypher_query` tool calls, starting from the early 3rd turn, indicates that the agent has already gathered enough information to start formulating and executing Cypher queries against the database thus validating the effectiveness of the designed toolset in facilitating an efficient **Formalisms Discovery**. From the 5th turn onwards, we can observe a rapid increase in `register_retrieval_result` tool calls, which indicates that the agent as already solved **Concepts-Formalisms Linking** thus starting to focus on organizing and compiling the results obtained from previous tool calls to construct the final result. Combining the figure 2c and the 2e, which shows a increasing trend of `register_retrieval_result` errors over LLM turns, we can infer that GPT-4.1 tend to struggle with effectively producing a valid final result due to the high complexity of the `register_retrieval_result` JSON Pydantic model that it must respect to successfully call the tool.

When we include failed chat sessions in the analysis (figure 2d), we can observe a similar trend until the 6th turn, after which there is a noticeable increase in `execute_cypher_query`, decrease in `register_retrieval_result` and a more minor predominance of the `get_properties_from` tool over all the other ones. Combining this observation with the figure 2f, which doesn't show a significant increase in `register_retrieval_result` errors over LLM turns, we can easily infer that for sure failed sessions were not caused by the inability of GPT-4.1 to correctly produce a valid JSON for the final result, but rather by the inherent complexity of some test instances that made the agent unable to successfully complete its execution within the imposed graph recursion limit or time constraints. This hypothesis, in fact, is further supported by the significant increase in `execute_cypher_query` tool calls from the 7th turn onwards without being followed by a corresponding increase in `execute_cypher_query` errors in figure 2f. It can also plausibly suggest that if the agent initially took a wrong path during the **Formalisms Discovery** phase, it could be forced to execute an high number of unnecessary Cypher queries to gather more information about the graph, thus quickly reaching the graph recursion limit or time constraints without being able to recover from its initial mistake. Thus this is a strong indication that LLMs cannot effectively be used to perform probabilistic or possibilistic algorithms that require backtracking capabilities to recover from initial mistakes. All these observations suggest that **Concepts-Formalisms Linking** is still a problem that cannot easily be solved by LLMs alone in fewer than 13 turns (as the maximum number of turns observed in figure 2b), thus requiring further research efforts to be effectively addressed to resolve more complex text-to-Cypher tasks.

Lastly, table 2 shows for each tool some statistics about the number of parallel tool calls, computed in a single LLM turn, averaged over all the tool call occurrences in all the chat sessions. From the mean and standard deviation we can easily confirm the fact that GPT-4.1 was able to exploit parallelism capabilities of our designed toolset to effectively reduce the overall number of LLM turns required to complete its execution.



(a) N. of Parallel Tool Calls Over LLM Turns

(b) N. of Chats Reaching Each LLM Turn

(c) Tool Calls Distribution of only Successful Sessions Over LLM Turns

(d) Tool Calls Distribution of all Sessions Over LLMS Turns

(e) Errors Distribution of only Successful Sessions Over LLM Turns

(f) Errors Distribution of all Sessions Over LLM Turns

Figure 2: Some Efficiency Statistics

Tool	Mean	Std	Mode	Min	Median	Max	Count
execute_cypher_query	1.01	0.10	1	1	1	4	4342
search_for	3.69	1.43	4	1	4	11	2614
get_properties_from	1.42	0.76	1	1	1	16	3813
get_domains_ranges_of	1.46	0.90	1	0	1	12	1924
get_components_with_property	2.05	1.47	1	1	2	9	523
register_retrieval_result	1.00	0.00	1	1	1	1	4659

Table 2
Tool Calls Statistics

5. Conclusions

In this paper, we presented an agentic approach to the Text-to-Cypher task that effectively bridges the gap between natural language questions and formal database queries. By decomposing the problem into distinct sub-tasks (1, 2, 3 and 4 in section 3) and empowering an LLM with a specialized suite of retrieval and validation tools, our system demonstrates that extensive fine-tuning is not a prerequisite for high performance in complex semantic parsing domains.

Our experimental results on the neo4j/text2cypher-2024v1 benchmark show that our zero-shot GPT-4.1-based agent achieves an Exact Match score of 32.62%, effectively matching the performance of a fine-tuned GPT-4o baseline (32.50%). Furthermore, the analysis of tool usage patterns reveals that the agent successfully adopts a "search-then-refine" strategy, leveraging high parallelism in early stages to broadly explore the schema before narrowing down to specific property and relationship constraints.

However, our analysis also highlighted limitations in current LLM-based agents, particularly their struggle with complex queries requiring backtracking or long reasoning chains, as evidenced by the correlation between failed sessions and high turn counts. Future work will focus on integrating more robust mechanisms that allow for non-linear reasoning and error recovery, as well as fine-tuning LLMs to reduce formatting errors and to better inject algorithmic behaviours. Ultimately, this work establishes a strong foundation for future research into autonomous database agents, suggesting that the combination of powerful general-purpose reasoning with domain-specific distinct tools is a promising direction for the next generation of Text-to-Query systems.

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